

Using the maXTouch Debug Port

1.0 INTRODUCTION

This application note describes the debug port on the maXTouch touchscreen controllers. These devices conform to the object protocol defined in the *Protocol Guide* for your maXTouch device. The *Protocol Guide* should be read in conjunction with this application note.

2.0 THE DEBUG PORT

2.1 Introduction

When the debug port is enabled, a one-directional flow of low level data is sent to the host. Most of this debug data is also available via the normal communications interface, but at a much lower data rate, thus limiting real time observation of signals.

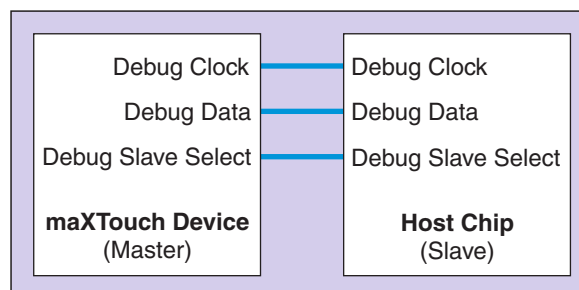
NOTE It can take a considerable amount of time to transmit the debug data over the debug port, so expect the device to operate substantially slower in this mode. Delays of tens of milliseconds can be expected in every cycle. This is normal.

The debug data transmitted by the debug port is typically used during the development of a touch-sensing application. It provides low-level information that is of use in tuning and refining the touch-sensing parameters and the system layout.

2.2 SPI Debug Port

The SPI debug port operates over the SPI interface (see [Figure 1](#)). The maXTouch device acts as an SPI master that generates data for analysis by an SPI slave. The slave is typically the host chip in a system in which the maXTouch device is a peripheral.

FIGURE 1: SPI DEBUG PORT



The SPI debug port comprises two main pins: **Debug Data** and **Debug Clock** plus, on devices that support it, a **Debug Slave Select** pin. Refer to the datasheet for your device for the location of the pins.

NOTE On some maXTouch devices, the pins used as the debug port are reassigned GPIO pins. Where this is the case, these pins cannot be used as GPIO pins.

2.3 USB Debug Port

Some maXTouch devices use the USB interface to provide debug monitoring via a USB Generic HID. A command is sent to the device and debug-monitoring data packets are returned using the debug port via the USB interface. Note that the **DEBUGCTRL** and **DEBUGCTRL2** fields in the Command Processor T6 must still be used to select which data will be sent (see [Section 2.4 "Generating Debug Data"](#)).

2.4 Generating Debug Data

The maXTouch devices can be commanded to generate debug data via the Command Processor T6 configuration object (refer to the *Protocol Guide* for your device for more information on this object).

Bytes 4 and 6 of the Command Processor T6 object are the debug controls (DEBUGCTRL and DEBUGCTRL2). By writing to these fields, the host chip can control the generation of debug data for the chip. These fields have the format shown in [Table 1](#).

TABLE 1: FORMAT OF COMMAND PROCESSOR T6 (GEN_COMMANDPROCESSOR_T6)

Byte	Field	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0 – 3	Refer to your <i>Protocol Guide</i>								
4	DEBUGCTRL	Reserved		SIGNAL	REFS	DELTAS	DELTAS8	TSCRN	CHIP
5	Refer to your <i>Protocol Guide</i>								
6	DEBUGCTRL2	DBGOBJMODEEN	Reserved						

DEBUGCTRL Field

SIGNAL: Enables the sending of each node signal value.

REFS: Enables the sending of each node reference value.

DELTAS: Enables the sending of each node delta value.

DELTAS8: Enables the sending of each node delta value in 8-bit format.

TSCRN: Enables the sending of the touchscreen data (for example, XY positions, states, and so on) for each supported touch.

CHIP: Enables the sending of the Command Processor T6 general status byte. Refer to your *Protocol Guide* for details of the status byte in the Command Processor T6 message data.

DEBUGCTRL2 Field

NOTE This field is available only on T2 and U series devices.

DBGOBJMODEEN: (Debug Object Mode Enable) This is a general control that applies to all objects that support Object Specific Debug Output. When Debug Object Mode is enabled, any of the objects that support Object Specific Debug Output can be enabled to output their own debug data. The format of the debug output will be specific for those objects. If multiple objects are enabled, the order of the data is product specific.

2.5 Format of the Debug Data

The data from the debug port is output in the data format given in [Table 2](#). The following sections give further information on the generated data.

TABLE 2: DEBUG DATA OUTPUT FORMAT

	Size (Bytes)	Description
Start Timestamp	1	Data Integrity – should match end timestamp
Chip Status	1	General Status
Touchscreen Status Touch[0]	19	Touch 0 Status Data
...	...	...
Touchscreen Status Touch[num_touches-1]	19	Touch num_touches-1 Status Data
Deltas 8-bit[0]	1 (Signed)	Node 0 8-bit Delta
...	...	...
Deltas 8-bit[num_nodes-1]	1 (Signed)	Node num_nodes-1 8-bit Delta

TABLE 2: DEBUG DATA OUTPUT FORMAT (CONTINUED)

	Size (Bytes)	Description
Deltas 16-bit[0]	2 (Signed)	Node 0 Delta LSByte
		Node 0 Delta MSByte
...	...	...
Deltas 16-bit[num_nodes-1]	2 (Signed)	Node num_nodes-1 Delta LSByte
		Node num_nodes-1 Delta MSByte
References[0]	2 (Unsigned)	Node 0 Reference LSByte
		Node 0 Reference MSByte
...	...	...
References[num_nodes-1]	2 (Unsigned)	Node num_nodes-1 Reference LSByte
		Node num_nodes-1 Reference MSByte
Signals[0]	2 (Unsigned)	Node 0 Signal LSByte
		Node 0 Signal MSByte
...	...	...
Signals[num_nodes-1]	2 (Unsigned)	Node num_nodes-1 Signal LSByte
		Node num_nodes-1 Signal MSByte
End Timestamp	1	Data integrity – should match start timestamp

Note 1: num_nodes = number of nodes. This can be calculated from the chip's Information Block by multiplying Matrix X Size by Matrix Y Size. Refer to your *Protocol Guide* for more information on the Information Block.

2: num_touches = the number of touches to be reported. This is set by the NUMTCH field in the Multiple Touch Touchscreen T100 configuration object (refer to your *Protocol Guide* for more details).

2.5.1 TIMESTAMPS

Generated debug data is always preceded by, and followed by, a one-byte timestamp value. This is a data integrity check for the SPI slave. The start and end timestamps should have the same values.

2.5.2 CHIP STATUS

The chip status information is a one-byte field containing the STATUS field from the Command Processor T6 message data (refer to your *Protocol Guide*).

2.5.3 TOUCHSCREEN STATUS

The touchscreen status data comprises multiple fields, one per touch, as defined in [Table 3](#).

TABLE 3: TOUCHSCREEN STATUS FORMAT

Size In Bytes	Description
1	Status Data. This is the STATUS field from the Multiple Touch Touchscreen T100 object's message data.
2	X Position – Filtered
2	Y Position – Filtered
2	X Position – Unfiltered
2	Y Position – Unfiltered
10	Reserved

2.5.4 8-BIT DELTAS

There is an 8-bit delta reported for each node. Each 8-bit delta consists of a signed value that represents the least significant byte of the 16-bit delta for that node.

Values larger than 127 are reported as 127; values smaller than -128 are reported as -128.

2.5.5 DELTAS

There is a signed 16-bit delta reported for each node.

2.5.6 REFERENCES

There is an unsigned 16-bit reference value reported for each node.

2.5.7 SIGNALS

There is an unsigned 16-bit signal value reported for each node.

2.5.8 DEBUG DATA FORMAT

Debug data bytes are transmitted most significant bit first. 16-bit values are sent least significant byte first.

2.6 Example

As an example, consider the case of `DEBUGCTRL = 0x12`, which enables the reporting of the touchscreen status and reference values. If the `num_touches = 2` and `num_nodes = 100`, the reported debug data has the format in [Table 4](#).

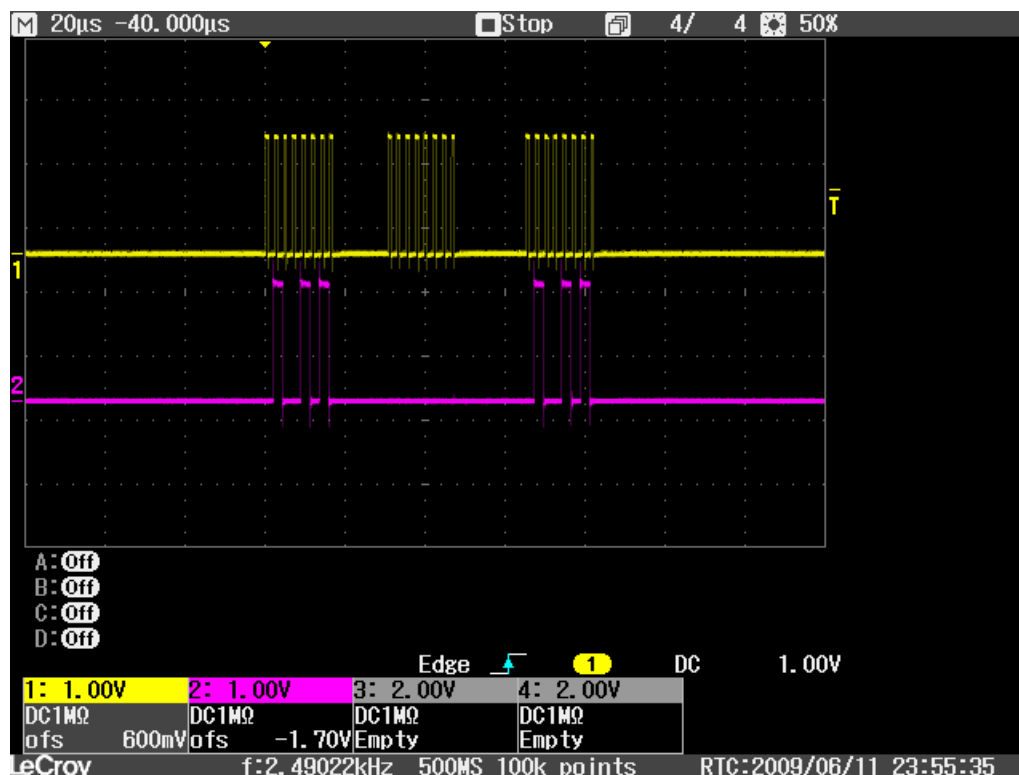
TABLE 4: EXAMPLE DEBUG DATA OUTPUT

Byte	Control Bit	Size (Bytes)
0	Start Timestamp	1
1–19	Touchscreen Status Touch[0]	19
20–38	Touchscreen Status Touch[1]	19
39–40	References[0]	2
41–42	References[1]	2
...	...	...
237–238	References[99]	2
239	End Timestamp	1

3.0 DEBUG DATA TIMING

3.1 Standard Devices

FIGURE 2: DEBUG PORT WAVEFORMS FOR CHIP STATUS BYTE



The scope shot in Figure 2 shows the debug port waveforms observed when the DEBUGCTRL field is configured to generate the chip status byte. As explained in Section 2.5 “Format of the Debug Data”, the data consists of a start timestamp byte, the chip status byte, and an end timestamp byte. In Figure 2 the upper (yellow) trace shows the debug clock line, and the lower (purple) trace shows the debug data line. The waveform timings are as in Table 5.

TABLE 5: TYPICAL WAVEFORM TIMINGS

Parameter	Value	Units
Debug clock high time	562	ns
Debug clock low time	576	ns
Debug data setup time	420	ns
Debug data hold time	786	ns
Gap between debug bytes	11.2	µs

3.2 Devices Supporting Slave Select

The scope shots in Figure 3 and Figure 4 show the debug port waveforms observed on a device running at normal data output speed when the DEBUGCTRL field is configured to generate the chip status byte. As explained in Section 2.5 “Format of the Debug Data”, the data consists of a start timestamp byte, the chip status byte, and an end timestamp byte. In Figure 3 the upper (yellow) trace shows the debug slave select line, the middle (green) trace shows the debug clock line, and the lower (blue) trace shows the debug data line. The waveform timings are as in Table 6.

FIGURE 3: NORMAL DEBUG DATA OUTPUT

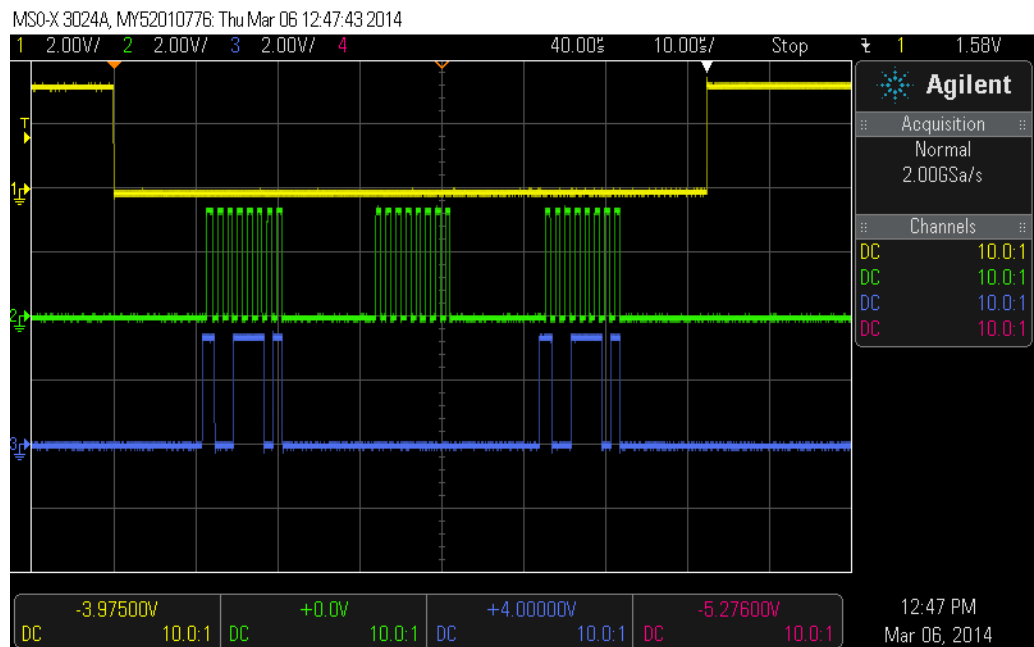
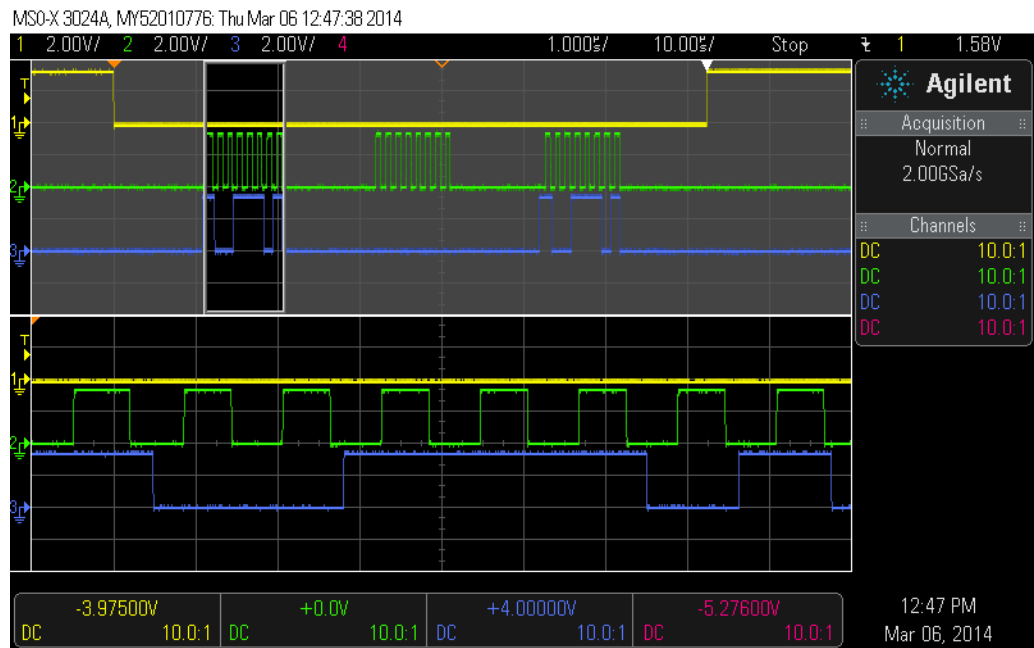


FIGURE 4: NORMAL DEBUG DATA OUTPUT – ZOOMED TO SHOW SETUP AND HOLD BEHAVIOR



The scope shots in [Figure 5](#) and [Figure 6](#) show the debug port waveforms observed on a device running at high speed data output when the DEBUGCTRL field is configured to generate the chip status byte. As explained in [Section 2.5 “Format of the Debug Data”](#), the data consists of a start timestamp byte, the chip status byte, and an end timestamp byte. In [Figure 5](#) the upper (yellow) trace shows the debug slave select line, the middle (green) trace shows the debug clock line, and the lower (blue) trace shows the debug data line. The waveform timings are as in [Table 6](#).

FIGURE 5: HIGH SPEED DEBUG DATA OUTPUT

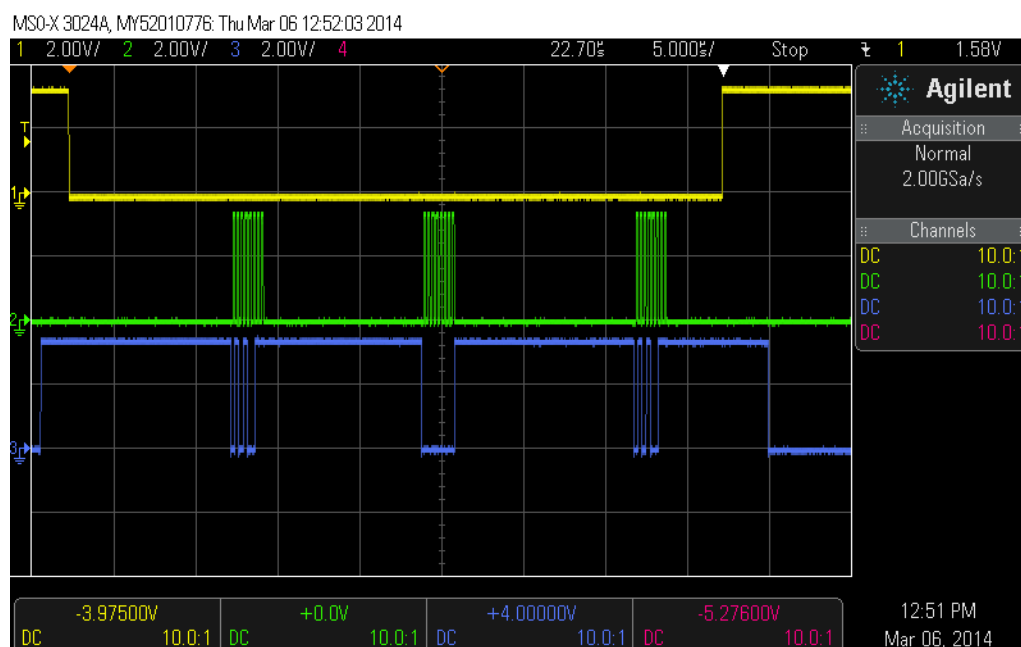


FIGURE 6: HIGH SPEED DEBUG DATA OUTPUT – ZOOMED TO SHOW SETUP AND HOLD BEHAVIOR

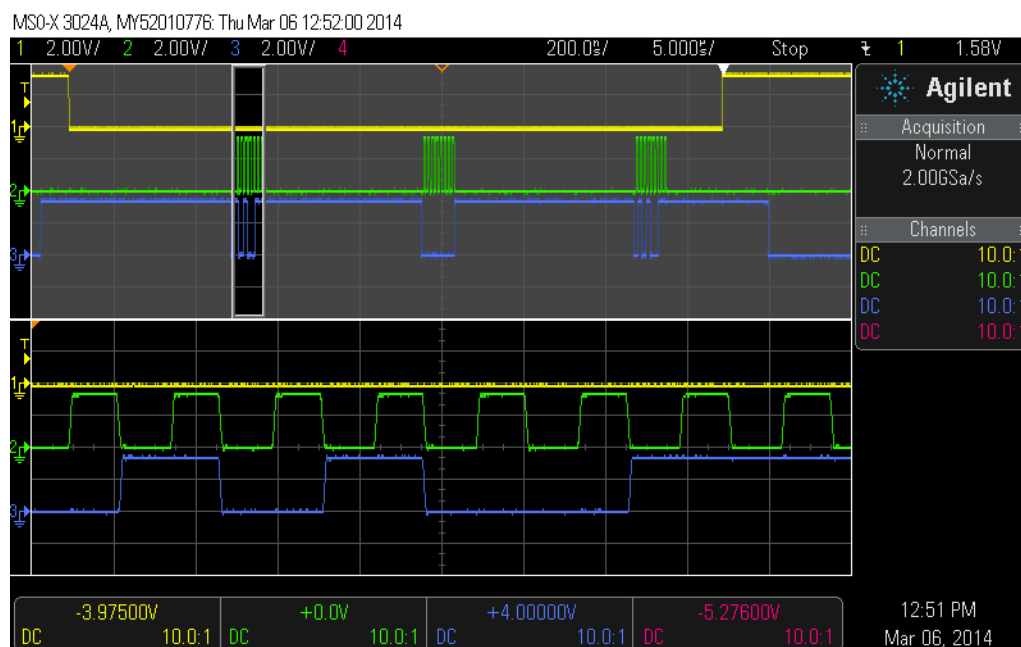


TABLE 6: TYPICAL WAVEFORM TIMINGS – DEVICES SUPPORTING SLAVE SELECT

Parameter	Normal Mode	High Speed Mode	Units
Debug clock high time	560	110	ns
Debug clock low time	570	120	ns
Debug data setup time	350	110	ns
Debug data hold time	650	110	ns
Gap between debug bytes	10	8	μs
SS assert to first clock time	10	9.5	μs
Last clock to SS de-assert time	10	2.7	μs

APPENDIX A: REVISION HISTORY

Revision DX (April 2014)

Final Atmel revision

Revision A (April 2017)

Reformatted and updated – Microchip edition

This revision incorporates the following updates:

- Updated to Microchip application note format:
 - [Table 5](#) updated
 - [Table 6](#) updated
 - Specific references to old devices removed
 - Revision History moved to this appendix
 - Back cover updated
- Changes to content:
 - Product features updated to remove mention of Product Data Store and passive stylus
 - [#Sect.Sensor]: restrictions on multiple Key Array layouts removed; no longer apply
 - References to Atmel Corporation removed or changed to Microchip Technology Inc, where appropriate
 - Object and parameter names updated to align with recent devices
- New documentation number assigned

NOTES:

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